PAPER_963: Buoyancy Gravity Triadic Mode

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Abstract

The Buoyancy Gravity Triadic mode evaluates $E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t) -$ threshold. Positive E_{net} drives expansion (nebulae, HII regions); negative E_{net} drives erosion (filaments, cometary knots).

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1. Net Energy

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} \cdot t) \cdot \exp(-\Gamma \cdot t) - \text{threshold}$$

2. Regime Classification

E_{net}	Regime	Astrophysical Example
> 0.1	Expansion	Nebulae, HII regions
< -0.1	Erosion	Filaments, pillars
[-0.1, 0.1]	Neutral	Transition zones

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)